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Structured-Visual Model for Dental Examination in Autism Spectrum Disorder Children: Cooperation and Compliance



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ABSTRACT

Introduction: Children diagnosed with autism spectrum disorder (ASD) may experience challenges in social interaction, communication, and sensory processing, which might influence their cooperative behavior during dental visits. To address this issue, visual pedagogy is commonly employed as a behavioral technique to facilitate successful dental examinations and treatments.

Objectives: The objective of this study was to evaluate the efficacy of a structured-visual behavioral approach in promoting compliance and cooperation during dental examinations in children with ASD in the Yogyakarta Special Region Province, Indonesia.

Methods: A structured visual study was performed at the Muhammadiyah University Dental Hospital during the 2021-2022 period. The study included 5 intervention visits at 1-week intervals and consisted of the Success Approach, Tell-Show-Feel-Do, Visual Pedagogy, Audiovisual Modeling, In vivo Modelling, Behavioral Trial, and Auto modeling. A total of 37 ASD children who met the inclusion criteria were included in the study. The study employed a quasi-experimental design comprising 1 pre-test and 3 post-tests. The efficacy assessment encompassed 2 key dimensions: firstly, the ability to comply with the stages of the dental examination, and secondly, the degree of cooperation was evaluated using the Frankl Behavior Scale (FBS).

Results: The findings showed that 64% of the cohort could achieve the highest score in achieving the highest score stage of dental examination in post-test 1. As for cooperation, 75% achieved a score of 4 (FBS) in post-test 1. Changes in the achievement of examination steps and cooperativeness after completing the intervention showed statistically significant changes with the Wilcoxon test ($P < .01$).

Conclusion: The structured-visual behavioral model approach was found to be effective in improving compliance and cooperation during dental examination in children with ASD in Yogyakarta Special Region Province, Indonesia.

Clinical relevance: These findings highlight the important point that the use of a structured-visual model in daily practice can be beneficial for children with ASD.

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Introduction

Autism spectrum disorder (ASD) is a neurodevelopmental disorder marked by deficits in 2 core domains: social interaction and communication and repetitive, restrictive, and stereotyped behaviors, interests, or activities.¹ ASD is a lifelong disability that the etiology is still unclear. Both genetic and environmental factors may contribute to the etiology.^{2,3} Comprehensive assessments can identify ASD symptoms within the first 3 years of a child's life.⁴

The global prevalence of ASD has increased in recent decades. According to the United Nations Educational, Scientific and Cultural Organization (2013), there were 35 million people with autism in 2011, which equates to 6 out of every 1,000 people worldwide with ASD. Approximately 1 in 68 children, or 1.5% of 8-year-olds, have been identified with ASD, according to the Autism and Developmental Disabilities Monitoring Network.⁵

Children with ASD often struggle with following instructions and communicating effectively and may be highly sensitive or insensitive to stimuli.⁶ These difficulties can make dental visits particularly challenging. Children may exhibit heightened behavioral distress, excessive anxiety, and uncooperative behavior during dental examinations and oral care, which can pose a significant challenge for dentists.⁷ It is important for dentists to be aware of these potential issues and to take steps to alleviate any discomfort or anxiety that the child may be experiencing.⁸

Previous studies investigated several factors that might cause children with ASD to be uncooperative during dental visits.^{9–11} Moreover, excessive fear of certain sounds, touch, or visual stimuli may play a role in cooperation.¹² This lack of cooperation can result in delays in essential dental care, as parents may be hesitant to take their children to the dentist if they exhibit uncooperative behavior.¹³ Therefore, children with ASD are at a higher risk of developing cavities due to poor dental hygiene, leading to potential periodontal diseases.^{14–16}

Several behavioral approaches, including sensory-adapted dental environments (SADE), desensitization, applied behavior analysis (ABA), Picture Exchange Communication System (PECS), Visual Schedule System (VSS), modeling, and Treatment and Education of Autistic and Related Communication-handicapped Children (TEACCH), have been used to reduce anxiety and uncooperative behavior during oral assessments and dental treatment.^{17–20}

The structured-visual behavior method commonly used with children with ASD is known as visual pedagogy. One of the methods is the TEACCH program, which focuses on visual elements and has shown effectiveness in addressing the traits of ASD.¹⁸ A rigorous meta-analysis has shown the substantial influence of TEACCH on the educational development of children with ASD.²¹ Research in Indonesia has explored the use of visual pedagogy in the dental examination for children with ASD.¹⁷ However, these investigations were not carried out in clinical settings with a large number of participants and prolonged monitoring assessments.

The hypothesis of the study is the structured-visual model of behavior approach effectively enhances the compliance

and cooperation of children with ASD during dental examinations. Therefore, this study aimed to evaluate the efficacy of the structured-visual behavioral approach in improving compliance and cooperation during dental examination for children with ASD in the Yogyakarta Special Region Province, Indonesia.

Methods

A quasi-experimental study was carried out to evaluate the application of a structured-visual behavioral model method and to analyze pre-and post-test results in dental examinations. The data collection was conducted from October 2021 to June 2022. The dental examinations and procedures were taken at a specialized pediatric dental clinic at Universitas Muhammadiyah Dental Hospital in Yogyakarta, Indonesia. In addition, the Health Research Ethics Committee of the Faculty of Medicine and Health Sciences, Universitas Muhammadiyah Yogyakarta, issued ethical clearance with approval number 256/EC-KEPK FKIK UMY/IX/2021. Additionally, the clinical trial was registered on ClinicalTrials.gov under the identifier NCT06470724.

Subject selection

The participants were selected from the Yogyakarta Autism Therapy Centre and specialized schools for children with ASD in the Yogyakarta Special Region Province, Indonesia. The inclusion criteria of this study were children aged between 6 and 18 years old (school-aged children in Indonesia), good general health without physical disabilities, cooperative parents or caregivers who provided informed consent to participate in the study, children residing in the same household as their parents or caregivers, and a diagnosis of ASD according to the Diagnostic and Statistical Manual of Mental Disorders, 5th Edition (DSM-5) with ASD with level 1 to level 3, by a psychiatrist. In addition, this study excluded children who had been trained for dental examinations at a dental health center, children who were declared by their parents and therapists as not amenable to non-pharmacological behavioral approaches in the dental examination, and those who could not comply with the study protocol.

Before starting the study, the sample size was determined by calculating the sample size formula. The formula was counted from the pre-and post-intervention data of dental compliance among children with ASD (score 1–10), as shown in a comparable previous study²² with an estimated mean difference of 2.0, a standard deviation of 2.3, and a power of 0.95. confidence level using G Power 3.1.9.7 software University of Dusseldorf Germany.²³ To compensate for dropout, 30% of the participants were added. Therefore, 24 participants were involved in this study. Additionally, with the allocated budget and high interest from parents to participate in this research, the study ultimately consisted of 37 individuals. Because the intervention applied in the study is an educational support program for dental check-ups, which provides significant benefits to the participants and does not harm them, this reason can be considered to address potential ethical questions that may arise.

Data collection

To evaluate the compliance with the dental examination and the level of cooperation exhibited by the participants during a dental examination, pre-test, and post-test measurements were conducted. One week prior to the scheduled appointment, the dental nurse communicated with the parents to notify them of the scheduled date and time.

The intervention process was conducted by the principal investigator, while a research assistant was responsible for data collection. The research assistant received training to accurately record the level of cooperation and compliance during the dental examination steps at the clinic according to the agreed procedures. The Cohen's Kappa coefficient was calculated to assess the accuracy in the evaluation of dental examination compliance (intra-examiner agreement: 0.9 [$P = .01$]; intra-examiner agreement: 0.90 [$P = .02$] in the FBS assessment. To ensure the absence of bias and the comfort of the children, the data collection process was conducted exclusively by one research assistant as an examiner who recorded the compliance of dental examination steps and the level of cooperation. In addition, the research assistant was blinded to the assessment process and was not involved in the analysis process to avoid the presence of bias.

A parent or caregiver accompanies the children to create a supportive environment in a dental setting. The outcomes of the measurements used in the study are listed below.

Compliance with the dental examination

Compliance with the steps of the dental examination consists of steps 1-10, each of which the researcher will mark as achieved. The measurement used was based on the previous study conducted by Orellana et al (2014).^{18,22}

The subject could enter the examination room when the call came in (score 1).

The child demonstrates a willingness to enter the examination room when their name is called.

Sitting down (score 2).

When the dentist stated, "Please sit down," the subject could sit comfortably in the dental chair and stay motionless for over 10 seconds.

Lying down in the dental chair (score 3).

The subject could lie back with legs flexed and head positioned on the headrest without assistance.

Enduring exposure to direct light on the face (score 4).

The subject could tolerate the dental light focused on the chest when the dentist said, "I will switch on the light, then I will move the light to your mouth," while the subject remained lying back.

Mouth wide open or showing the teeth (score 5).

The subject could open their mouth wide when the dentist said, "Open your mouth, please." This was conducted twice.

Tolerating touching of the mouth with gloves (score 6).

The subject allowed the dentist's finger to touch the teeth and the oral mucosa to remain open for at least 5 seconds when the dentist instructed them to "Open your mouth."

Examination with a mirror (score 7).

The subject tolerated the mouth mirror inside the oral cavity for at least 5 seconds when the dentist said, "I will count

your teeth." The children counted by themselves. This activity was performed twice.

Examination with a dental explorer (score 8).

The subject could tolerate the dental explorer inside the oral cavity.

Examination with a mirror and dental explorer (score 9).

The subject could tolerate the explorer inside the oral cavity for at least 5 seconds when the dentist said, "Open your mouth; I will clean your teeth with this stick." This activity was performed twice.

Dental occlusion (score 10).

The subject could make centric occlusion or press the upper and lower teeth together for at least 5-second intervals. This activity was performed twice. The dentist also asked them to show their teeth and started counting from 1 to 5.

Cooperation

The Frankl Behavior Scale (FBS) was employed to assess the level of cooperation exhibited by participants in both the pre- and post-tests. The Frankl Behavior Scale (FBS) is extensively utilized in clinical behavioral research to assess individual cooperativeness, owing to its validity and reliability.^{24,25}

The scale was partitioned into the following 4 categories^{26,27}:

Definitely negative (score 1).

Rejection of the treatment, crying, being fearful, struggling, and showing extreme negativism.

Negative (score 2).

Unwilling to accept dental treatment, non-compliant, showing negative attitudes, but not very pronounced.

Positive (score 3).

Exhibits high levels of tolerance toward any medical procedure, possesses a positive inclination and mindset toward dental care, diligently adheres to instructions, and actively cooperates.

Definitely positive (score 4).

Good rapport with the dentist, interested in dental procedures, smiles, and enjoys the situation.

Intervention

The duration of each intervention session ranged from 15-20 minutes, as scheduled by the nurse. Each participant must confirm their attendance so that the researcher can prepare the visit properly according to the intervention stage. The intervention was conducted for 5 weeks. Prior to the intervention, the research assistant assessed the achievement of the 10 steps of dental examination (a)-(j)) and cooperation with the Frankl Behavior Scale (FBS). During this activity, the dental nurse documented it on video. The intervention consisted of 7 behavioral approaches in this study, which were defined as follows:

Successive approach

The participants will be introduced to and interact with dental researchers, and a series of instruments will be utilized in the examination, as well as a step examination with a custom-made dummy inserted with a human tooth model.

Tell-Show-Feel-Do (T-S-F-D)

The dental examination is explained to the patient by providing a detailed account of the procedures, the patient's sensations and actions, and the completed tasks.

Visual pedagogy

The dental examination is demonstrated using visual media, such as pictures and photographs, to explain the steps of the dental examination.

Audiovisual modeling

The steps of the dental examination are presented with an educational approach that utilizes audiovisual resources, such as videos and engaging animations.

In-vivo modeling

Activities in dental examinations are demonstrated through observational learning by siblings or peers who have a cooperative attitude in examinations.

Behavioral trials

Participants are introduced to the actual situation of a dental examination room, interact with dental researchers, and are also introduced to a series of instruments that will be used in the examination. This process will make participants anticipate the examination steps they will receive.

Auto-modeling

The participants were shown a video in which they acted as video models from previous visits that demonstrated cooperative behavior. This video modeling technique was done after the video recordings of the previous visits were edited to show only the cooperative attitudes of the participants.

Before conducting the study, the researcher acquired research permits from local authorities and contacted schools and therapy centers. The researcher elucidated the objective and advantages of the study to the director of the therapy centers and the school principals while also providing the ethical clearance and research permission that had been granted. The educators and clinicians issued invitation letters to the parents to ascertain their willingness to participate in the study. Subsequently, the researcher extended an invitation to parents who had shown interest in taking part in the clinical intervention. Potential participants were contacted, and a meeting was arranged. Additionally, a timetable was established for the arrival of each participant. Actually, 46 parents responded by contacting the researcher to join, but only 45 submitted the informed consent.

In the initial visit, the principal investigator gathered data on participants' socio-demographics, medical history, dental health, severity level of ASD, communication skills, and language comprehension ability. Furthermore, in this study, the

research assistants commenced data collection on the children's ability to perform dental examination procedures and level of cooperation. This visit was a pre-test visit. Forty-five participants participated in the pre-visit study. The research flow is shown in [Figure 1](#) below:

At the first intervention visit, the successive approach ([Figure 2A](#)), the T-F-S-D method ([Figure 2B](#)), and visual pedagogy ([Figure 2C](#)) were employed. The researcher employed successive approach within the examination room. The children were introduced to the clinic environment, the dentist or researcher, the assistants, the dental staff, the dental chairs, and the basic instruments. The children are provided with the opportunity to engage with and utilize dental instruments and, dental models. Moreover, the researcher employed the use of puppets as a pedagogical tool to facilitate comprehension and guide the subjects through each step of the dental procedure ([Figure 2C](#)). This guarantees their active involvement and comprehension. The participants were provided with clear and concise verbal explanations of each stage of the oral examination, with the assistance of parents or caregivers. Subjects were provided with photographic representations ([Figure 2C](#)) of the dental examination procedure. Prior to departing the clinic, the dentist provided the parents or caregivers with a set of images depicting the dental examination procedure ([Figure 2C](#)). In the evening, the parent or caregiver was suggested to review the dental visit by displaying the images.

In the second intervention visit, a variety of pedagogical approaches were employed, including successive approach ([Figure 2. A](#)), T-S-F-D ([Figure 2B](#)), visual pedagogy ([Figure 2C](#)), and audio-visual modeling ([Figure 2D](#)). Tablets or computers were utilized for audio-visual modeling, presenting a video of the 10 steps of the dental examination. The dentist encouraged the participants to perform the activity on a dummy while watching the dental examination video on a smartphone or other multimedia device, allowing them to see and hear each step. At home, parents or caregivers were advised to show photographs and videos from the clinic to reinforce the dental exam steps.

During the third dental visit, the dentist used successive approaches ([Figure 2A](#)), T-F-S-D ([Figure 2B](#)), visual pedagogy ([Figure 2C](#)), and in-vivo modeling ([Figure 2E](#)). Strategies in-vivo modeling allowed the participants to focus their gaze on a model. The model can be a peer who is cooperative in the dental examination and is able to perform all the necessary procedures. The parents or caregiver at home again showed the photographs and interacted with the child to repeat the dental examination to refresh their memory of the examination process.

In the fourth intervention visit, the researcher used various approaches such as successive approach ([Figure 2A](#)), T-F-S-D ([Figure 2B](#)), visual pedagogy ([Figure 2C](#)), and behavioral trials ([Figure 2F](#)). At this visit, the participant started to try out the ten steps of dental examination ([Figure 2F](#)). The parent or caregiver was again given instructions to perform the ten steps of the examination with the subject at home while participating in role-play activities, including the parent or caregiver and siblings. A plastic diagnostic device is used to simulate the process. This made the dental examination procedure easier to remember.

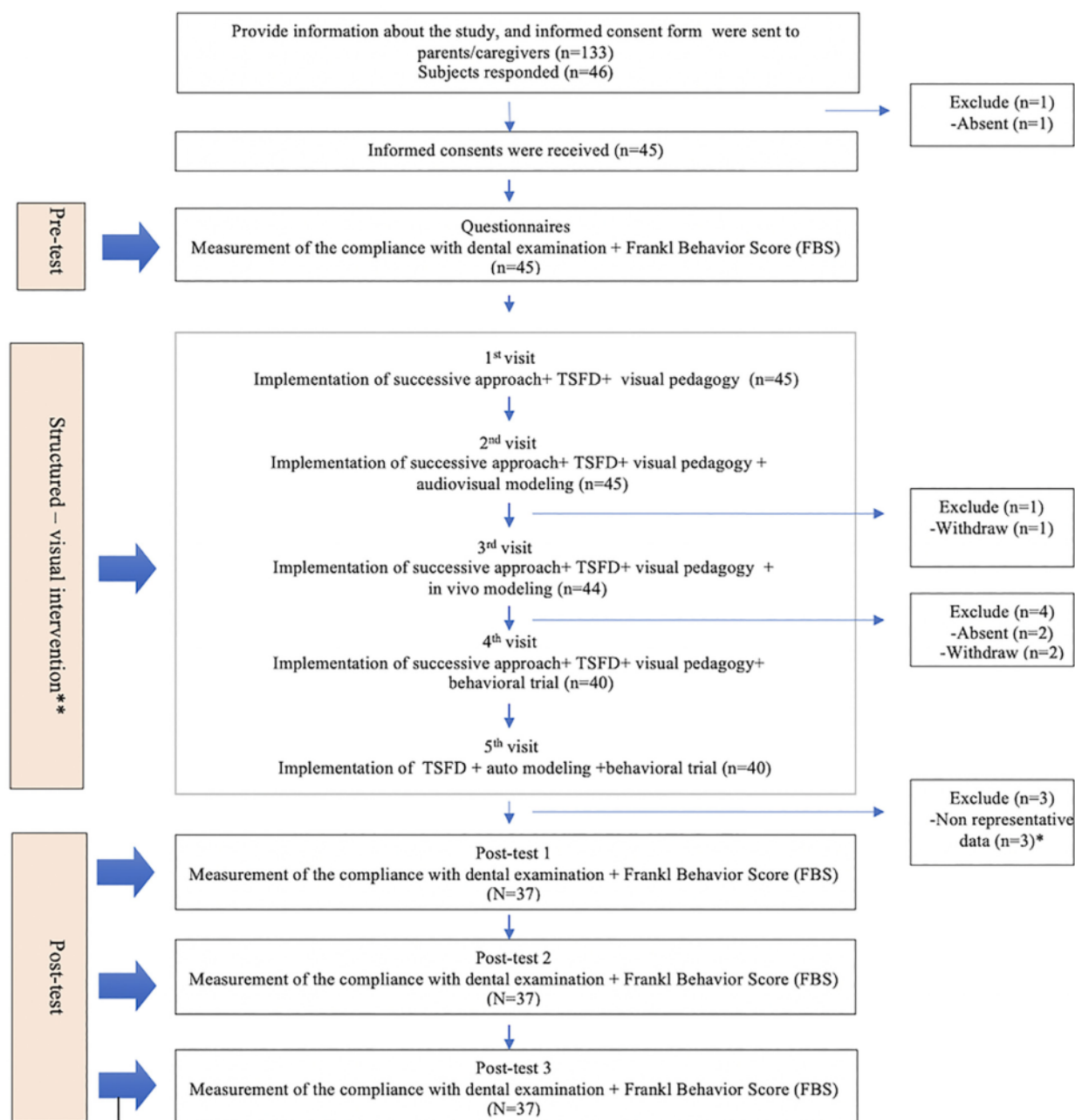


Fig. 1 – Research methodology flowchart.

At the fifth intervention visit, which was the final visit in the series of interventions, the dentist employed the T-S-F-D (Figure 2B), behavioral trials (Figure 2F), and auto modeling (Figure 2G). The objective of auto-modeling is to reinforce the child's success in exhibiting positive behavior. The dentist presents edited visual materials, including photographs and videos, from previous dental visits that illustrate the child's cooperation. On the night before the next scheduled visit, the parents showed the pictures of the dental examination step again to recall the child's memory of the dental examination.

The initial post-test was administered 1 week following the intervention, followed by a subsequent post-test 2 weeks

later, and a final post-test 1 month after the second post-test. The researchers gathered data on compliance with dental steps and cooperation throughout these examinations. If the subject failed to attend the clinic as scheduled, the subject was obligated to proceed with the intervention measures during the subsequent visit, which followed the set timetable in a sequential manner.

Statistical analysis

Data were analyzed using Statistical Package for Social Science Version 20.0 software (SPSS 26.0, SPSS Inc., Chicago, IL, USA), with a 95% confidence level. The Wilcoxon signed-rank



Fig. 2 – (A) Successive approach (B) Tell-feel-show-do (T-F-S-D) (C) Visual pedagogy (D) Audio-visual modeling (E) In-vivo modeling (F) Behavior trials (G) Auto modeling.

test was used to compare the results obtained from the pre and post-test on the dependent variables: dental examination compliance and cooperative attitude. Friedman test results were used to show the comparative results of the Wilcoxon test in 4 assessments. Statistical significance is indicated by a value of $P < .05$.

Results

One hundred thirty-three ASD children were invited to participate in this clinical research study. Only 45 individuals registered for the initial trial. Initially, 45 participants attended the first intervention visit through the second intervention visit, however, at the fourth intervention visit, 2 participants withdrew again, and by the end of the intervention phase, 2 more participants withdrew. Therefore, only 40 participants remained committed to following the study until the completion of the third post-test. In the process of data analysis, 3 participants were

identified as outliers, resulting in the decision to present data only from 37 individuals in this study.

Table 1 presents the descriptive data regarding the subjects involved in this study. Most of them were male, with a male-to-female ratio of 3.6:1. The age group ranging from 6 to 8 years exhibited greater dominance than the other age groups. In this study, psychiatrists assessed the severity of ASD and determined that the majority of participants had mild (40.5%), moderate (37.8%), and severe (21.6%) levels of ASD, respectively.

Furthermore, this study also shows that there is no statistically significant association between gender, age group, and severity level of ASD with dental check-up compliance or cooperativeness analyzed by the chi-square test and Fisher's Exact test ($P > .05$). This study also examined several factors influencing dental examination compliance and cooperation, including communication skills, language comprehension, and prior dental visit experience. The findings indicated that the ability to comprehend 1-way directions, as an aspect of communication abilities, is the only

Table 1 – Descriptive information of the subjects.

Variable	N (%)	Compliance of dental procedure	Cooperation
P-value			
Gender			
Male	29 (78.4)	0.30	0.24 [†]
Female	8 (21.6)		
Age (years old)			
6-8	12 (32.4)	0.572	0.89 [†]
9-11	9 (24.3)		
12-14	7 (18.9)		
15-18	9 (24.3)		
Severity			
Mild	15 (40.5)	0.125	0.3
Moderate	14 (37.8)		
Severe	8 (21.6)		
Communication ability			
One way communication			
Able	32 (86.5)	0.005*	0.34
Unable	5 (13.5)		
Picture			
Able	14 (37.8)	0.133	0.08
Unable	23 (62.2)		
Sign Language			
Able	21 (56.8)	0.702	0.66
Unable	16 (43.2)		
Language comprehension			
Able	9 (24.3)	0.731	0.05 [†]
Unable	28 (75.7)		
Dental visit experience			
Yes	25 (67.56)	0.743	0.713
No	12 (32.43)		

Pearson Chi-square.

* Significant P-value < .05.

[†] Fisher's Exact Test.

statistically significant factor influencing compliance to the step dental examination.

Compliance with dental step examination

The average compliance score achievement in the pre-test was 6.81 ± 3.125 , the mean in post-test 1 that showed an increase after the intervention was 9.10 ± 1.83 , and the means in post-tests 2 and 3 were 9.24 ± 1.73 and 9.54 ± 1.53 , respectively.

Post-test 1 indicates that 24 subjects reached step 10 (the highest score), and 2 subjects showed regression (Numbers 35 and 37), as shown in Figure 3 (a). Meanwhile, as shown in Figure 3 (b), the results in post-test 2 showed that most subjects achieved better dental examination steps in the assessment. Two subjects showed regression (13 and 35). The results of post-test 3, as exhibited in Figure 3 (c), demonstrated that almost all subjects achieved the highest score at this stage (10). However, some subjects only achieved a score of 8; they were subjects 2 and 6, who achieved a score of 7. Subjects who experienced a decrease in performance were subjects 13 and 35.

The results generally present the assessments made by the 37 subjects to achieve the dental examination steps. Thirty-three out of 37 subjects achieved the highest scale (score 10) in post-test 3 (89%), 26 subjects (70%) showed improvement from the pre-test, and only 2 subjects (5%) had a decrease in achievement. The Wilcoxon signed-rank test was used to assess the importance of changes in compliance in the dental steps and cooperation before and after the intervention (pre- and post-test). Table 2 displays the test results that demonstrated the comparative compliance outcomes of the Wilcoxon test in the 4 assessments. The statistical analyses yielded significant results ($P < .001$) for the comparisons between the pre-test and the post-test 1, post-test 2, and post-test 3, respectively. Nevertheless, the examination of the results from post-tests 2 and 3 did not reveal a statistically significant difference ($P > .05$).

Cooperation

Figure 4 shows the comparison results between pre-test and post-test 1, post-test 2, and post-test 3 regarding the level of cooperation in visiting the dentist for each subject. Thirty or more subjects showed remarkable achievements in terms of cooperation in post-tests 1, 2, and 3 after the structured-visual model intervention was implemented.

In general, there was an improvement in cooperation. More than 75% in post-test 1, as shown in Figure 4(a), showed a positive change (a score of 4); Figure 4 (b) exhibits that most subjects' cooperation also improved in post-test 2. In contrast, subject 37 experienced a decrease in performance since the previous test. Another piece of additional information is that subject 13 showed a stagnation in performance at an uncooperative level after the intervention, while subject 15 showed an increase, although still at an uncooperative level. Finally, in post-test 3, as shown in Figure 4 (c), 89.1% achieved a level of cooperation of 4 in the post-test, revealing that the majority of individuals exhibited noteworthy enhancement in terms of cooperation. The majority of subjects attained the highest score, with the exception of subjects number 2, 15, and 35, who achieved a score of 3 but were previously classified in the cooperative category. Regrettably, subject number 13 exhibited a consistent lack of cooperation, remaining at level 2 and displaying continued uncooperativeness.

Furthermore, Table 2 shows significant values of $P < .05$ in each assessment compared to the pre-test, except between post-tests 2 and 3, in which the change is no longer significant. The average achievement of the Frankl Behavior Scale (FBS) was 2.45 ± 0.76 in pre-test, 3.59 ± 0.68 in post-test 1, 3.70 ± 0.61 in post-test 2, and 3.86 ± 0.41 in post-test 3.

These results demonstrate a significant improvement of cooperation in dental examination achievement in the 3 subsequent examinations compared to the initial pre-test. Regarding the Frankl Behavior Scale, it is important to note that 36 out of 37 participants (97%) reached the cooperative stage.

Discussion

The objective of this study was to evaluate the efficacy of the structured-visual approach in promoting compliance and

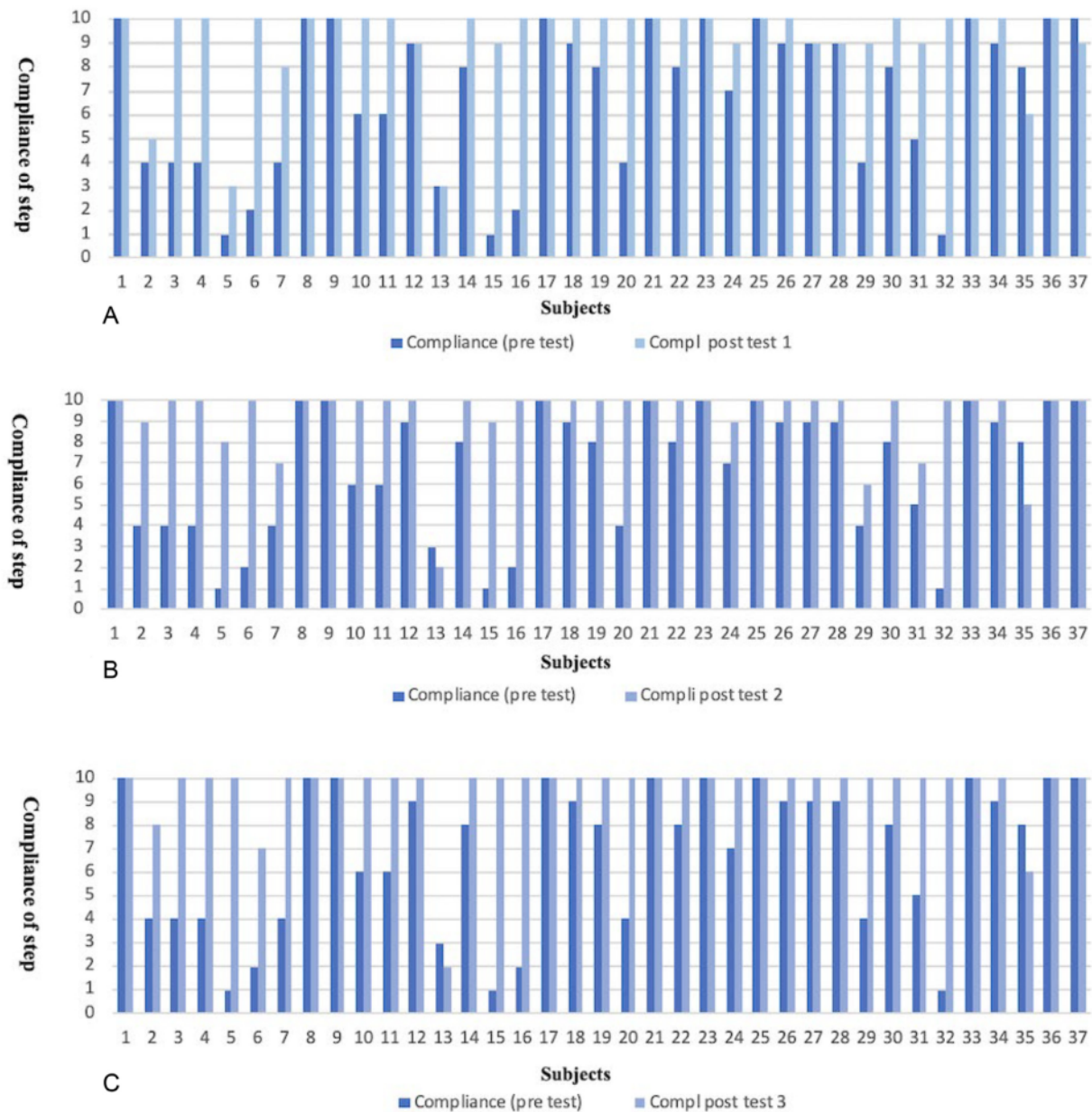


Fig. 3 – The results of analyzing the compliance of dental step examinations. (A) Comparison of the Compliance of Dental Examination between Pre-Test and Post-test 1. (B) Comparison of the Compliance of Dental Examination between Pre-Test and Post-test 2. (C) Comparison of the Compliance of Dental Examination between Pre-Test and Post-test 3.

cooperation during dental examinations of children with ASD.

This finding indicated an improvement in compliance to the dental examination protocol among the participants, resulting in most of them achieving the maximum score of 10 in post-test 3. Nevertheless, some individuals were unable to achieve the highest score or experienced a decline in performance on post-test 3 compared to their pre-test findings. It is noteworthy that participants who achieved the highest score of 10 on the compliance section of the pre-test exhibited a decline in their performance on the post-test, a deviation from the expected outcome.²⁹

The intervention's efficacy in improving cooperation skills was demonstrated by the notable disparities in the FBS scores prior to and following post-tests 1, 2, and 3. Furthermore, 89.1% of the subjects attained the highest FBS score, indicating a favorable outcome with a score of 4. In addition, this intervention led to a 54% increase in the number of participants who shifted from the non-cooperative behavior levels 1-2 to the cooperative collaboration levels 3-4.³⁰

Acceptance of the provided sensory input can be determined during the intervention stage. At the dental examination stage, sensitivity to light can be determined by the child's ability to tolerate the light emitted from the dental chair lamp (step 4 and 5), response to tactile stimulation

Table 2 – Friedman test result of compliance with dental step examination and cooperation.

Assessment	Mean $\pm$ SD	Median (min-max)	P-value
Compliance with dental examination			
Pre-Test	6.81 $\pm$ 3.12	8 (1-10)	
Post-test 1	9.10 $\pm$ 1.83	10 (3-10)	<.001*
Post-test 2	9.24 $\pm$ 1.73	10 (2-10)	<.001*
Post-test 3	9.54 $\pm$ 1.53	10 (2-10)	<.001*
Post-test 2-Post-test 3			.12
Cooperation(FBS)			
Pre-Test	2.45 $\pm$ 0.76	2 (1-4)	
Post-test 1	3.59 $\pm$ 0.68	4 (2-4)	<.001*
Post-test 2	3.70 $\pm$ 0.61	4 (2-4)	<.001*
Post Test 3	3.86 $\pm$ 0.41	4 (2-4)	<.001*
Post-test 2- post-test 3			.58

* Significant, *p*-value <0.05 compared to the pre-test. at post-hoc Wilcoxon. FBS, Frankl Behavior Scale.

when the dentist touches the mouth with gloves (step 6), and response to taste when the mouth mirror and explorer are used to examine the mouth (step 7, 8, and 9). In pre-test 1, 1 subject (subject 13) could not tolerate the light produced by the dental unit. Until the end of the assessment, this subject exhibited difficulty in tolerating the sensory output of the dental unit light. Moreover, children with ASD who are over-responsive to certain sensory stimuli can gradually undergo sensory recognition training for fear-provoking stimuli in the dental clinic, including noise, the bright light from a dental unit, mouth touching, and unfamiliar tastes or textures.^{28,29}

The analysis of compliance in the pre-test and post-tests 1, 2, and 3 revealed significant *p*-values for compliance with exam procedures and collaborative behavior. However, the researcher did not obtain consistent results regarding achievement between post-tests 2 and 3. This discrepancy could be attributed to several subjects achieving the maximum score in post-test 2.

The structured-visual behavior approach has proven effective in enhancing compliance and cooperation in children with ASD during dental examinations. The improvement in cooperation and step compliance in dental examinations following the structured-visual approach was in line with previous studies.^{18,30} Additionally, children with ASD are more likely to live according to the same order, schedule, and regularity to which they are accustomed. Hence, this visual-structured intervention was organized with scheduled visits at a specific time in the same place.³¹ Most subjects looked forward to “dental check-up time” every week and showed less anxiety than during the first visit. Previous studies have revealed that visual learning as part of visual pedagogy is the preferred learning method in populations with ASD.^{32,33}

Pictures, videos, modeling, and trials can introduce the recognition of strange smells, sounds, tastes, and environments to overcome challenging behaviors. In addition, the peer modeling and self-video modeling applied in this study help subjects learn and strengthen their motivation to reach the stage of better cooperation.^{34,35}

During the study, the parents were instructed to repeat the processes performed in the clinic to familiarize themselves with sensory perception.³⁶ According to previous reports, the interaction process between parents and children with ASD can improve children's emotional self-regulation.³⁷ The role of parents in building trust and confidence in visiting the dentist also contributes to the formation of the subject's behavior toward dental examinations.³⁸ Under these circumstances, dentists must be able to convince parents to restrain such desires and to consider the child's situation, which may lead to increased anxiety or deterioration of the subject's cooperative attitude. Another challenge was when the subjects needed to express their discomfort during the examination but failed due to a disability in communication, which led to the subjects' anger and contributed to uncooperative behavior during the dental examination. Some parents could cope with this situation and encouraged the subject to continue treatment, but most of them needed to reschedule the visit due to uncondusive emotions.

Introducing routine dental examinations with a structured-visual approach both in the dental clinic and at home could reduce stress, maintain an excellent professional-patient relationship, cost less, minimize referrals for treatment under general anesthesia, and establish positive communication with children. In addition, visiting the dental clinic for those with challenging behaviors requires meticulous preparation. The previous literature stated that parents would appreciate it if dentists listened to them and knew the children's needs to prepare for the best appointment.³⁸ Dentists have suggested asking parents about the frequency, triggers, and ways of coping with challenging behaviors. Therefore, the role of parents and caregivers in communicating current emotions and recent activities is essential information for researchers to interpret.

The calculation of sample size according to the Frankl Behavior Scale as the secondary outcome brings to the issue that the sample size was smaller than the compliance or first outcome, and the ability the power to detect the difference using the Frankl Behavior Scale might be questioned. This study utilized a compliance outcome sample size, which is greater than the calculated sample size for FBS. This larger sample size was chosen to ensure adequate power for detecting differences.³⁹ Additionally, the participants in this study ultimately numbered 37 because some parents expressed interest in enrolling their children in the clinical study, which exceeded the calculated sample size, prompting the researchers to add more participants. According to Murti (2016), a larger sample size improves the research's ability to detect relationships between variables (statistical power), reduces the role of chance factors in concluding variable relationships, and minimizes the possibility of sampling error and standard error.⁴⁰

The external validity of this study is determined by the extent to which the structured visual approach can be generalized to children with ASD in dental clinics equipped with comparable equipment, facilitating accomplished dental examinations and enhanced cooperativeness as measured by FBS. However, the researcher realized that interventions for diverse subjects resulted in different acceptance levels. The variability in the ability of the subjects to cooperate and

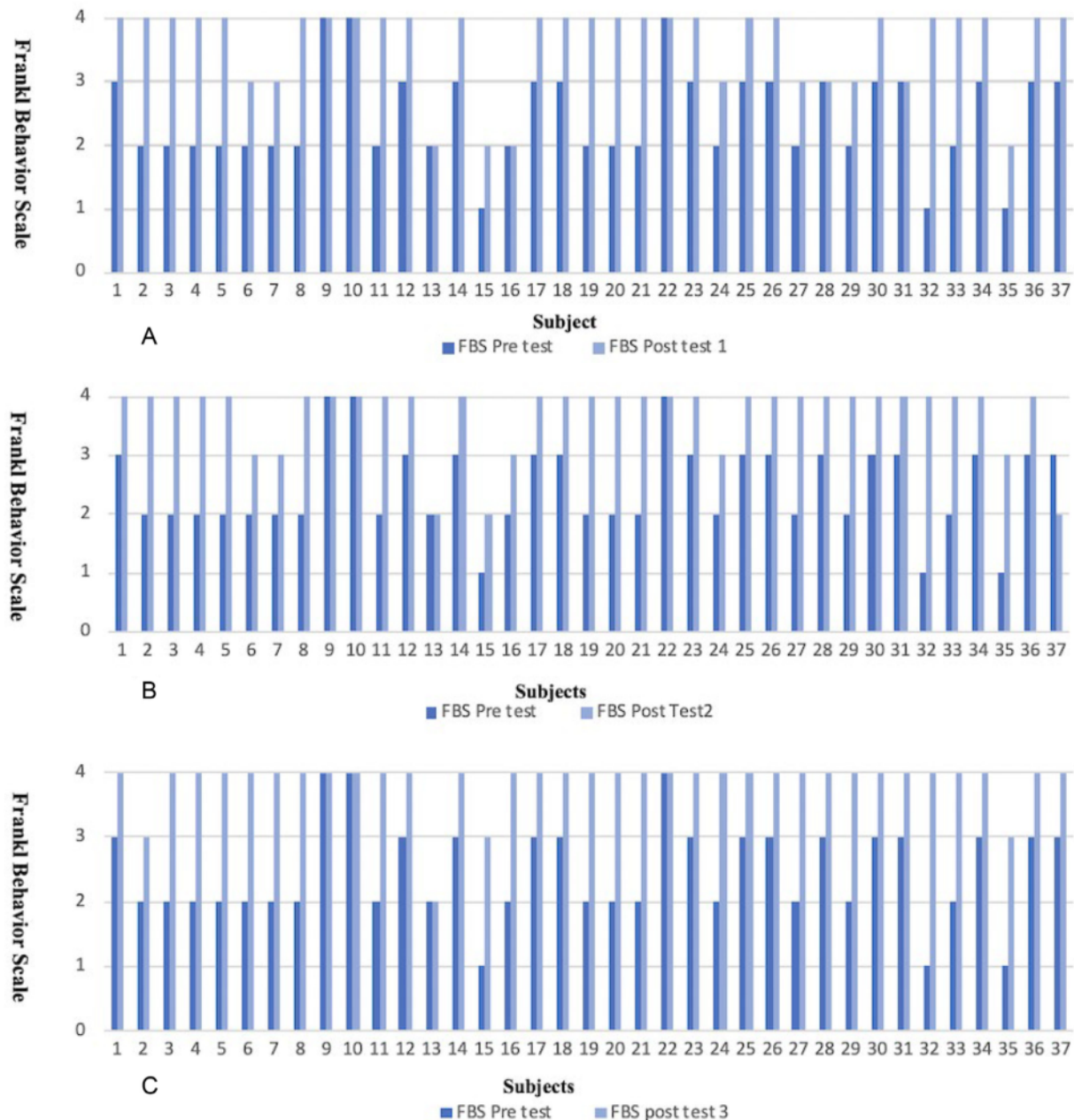


Fig. 4– Comparison of the Frankl Behavior Scale in the assessments. (A) Comparison of the Level of Cooperation between Pre-Test and Post-Test 1. (B) Comparison of the Level of Cooperation between Pre-Test and Post-Test 2. (C) Comparison of the Level of Cooperation between Pre-Test and Post-Test 3.

interact caused this outcome. In addition, some circumstances need to be highlighted that might have required postponing a visit to the dental clinic because if the intervention or assessment had been carried out, it would not have worked well or produced biased test results. These circumstances include whether the subject was emotionally unstable, irritable, hyperactive due to the administration of certain nutrients or drugs, or excessively anxious about new things to be taught.

The main limitation of this study is the absence of a control group in the intervention due to various characteristics within the ASD population. The sampling method employed

is purposive and utilizes non-random sampling. The principal investigator and data collector are 2 individuals who consistently interact with the participants, which may result in potential bias. To minimize biases, the researcher implemented thorough training for data collectors to stick to the study protocol, using standardized measurement outcomes. Furthermore, the study employed blinding for the data collector, ensuring that the collector remained uninformed about the assessments, whether pre-test or post-test 1,2,3, and finally implemented a rigorous follow-up protocol to ensure comprehensive data collection.⁴¹ The recommendation for future studies is to include a control group that accurately

reflects the characteristics of the population whenever feasible and to employ randomization for allocating participants within similar groups to mitigate any bias in the research outcomes.

In conclusion, this study demonstrates the significant effectiveness of a structured- visual behavioral model in improving compliance and cooperation of children with ASD during dental examinations in Yogyakarta Special Region Province. The application of this model to a population of school-aged ASD children in Yogyakarta Special Region Province, Indonesia, resulted in a marked improvement in compliance with dental examination procedures and cooperation levels. These findings underscore the potential of the structured-visual model as a valuable approach in pediatric dental care for children with ASD, advocating for its wider application in similar clinical settings to elicit cooperative behavior.

Conflict of interest

The authors declare no potential conflicts of interest with respect to the authorship and/or publication of this article.

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Authors' contributions

1. Alfini Octavia conceptualized the study and collected and analyse the data.
2. Thanya Sitthisetpong contributed to conceptualizing the study, revising it critically for important intellectual content.
3. Matana Kettaratad-Pruksapong contributed to conceptualizing the study, revising it critically for important intellectual content.
4. Iwan Dewanto participated in manuscript writing.

All authors reviewed and approved the final version of the manuscript.

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REFERENCES

1. American Psychiatric Association. Supplement To Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition [Internet]. Diagnostic and Statistical Manual of Mental Disorders; 2015. [cited 2024 Mar 20]. Available from <http://dsm.psychiatryonline.org>.
2. Sauer AK, Stanton JE, Hans S, Grabrucker AM. Autism spectrum disorders: etiology and pathology. *Autism Spectrum Disorders* [Internet]. Brisbane: Exon Publications; 2021. p. 116. doi: [10.36255/exonpublications.autismspectrumdisorders.2021.etiology](https://doi.org/10.36255/exonpublications.autismspectrumdisorders.2021.etiology).
3. Havdahl A, Niarchou M, Starnawska A, Uddin M, Van Der Merwe C, Warrier V. Genetic contributions to autism spectrum disorder. Vol. 51, *Psychological Medicine*. Cambridge University Press; 2021. p. 2260–73. <https://doi.org/10.1017/s0033291721000192>
4. Ellerbeck K, Smith C, Courtemanche A. Care of children with autism spectrum disorder. *Primary Care - Clinics in Office Practice* [Internet] 2015;42(1):85–98. doi: [10.1016/j.pop.2014.09.004](https://doi.org/10.1016/j.pop.2014.09.004).
5. Kogan MD, Vladutiu CJ, Schieve LA, et al. The prevalence of parent-reported autism spectrum disorder among US children. *Pediatrics* 2018;142(6):1–11. doi: [10.1542/peds.2017-4161](https://doi.org/10.1542/peds.2017-4161).
6. Kuhaneck HM, Chisholm EC. Improving dental visits for individuals with autism spectrum disorders through an understanding of sensory processing. *Spec Care Dentist* [Internet] 2012;32(6):229–33. doi: [10.1111/j.1754-4505.2012.00283.x](https://doi.org/10.1111/j.1754-4505.2012.00283.x).
7. Pastore I, Bedin E, Marzari G, Bassi F, Gallo C, Mucignat-Caretta C. Behavioral guidance for improving dental care in autistic spectrum disorders. *Front Psychiatry* [Internet] 2023;14:1–12. doi: [10.3389/fpsy.2023.1272638](https://doi.org/10.3389/fpsy.2023.1272638).
8. Stein LI, Lane CJ, Williams ME, Dawson ME, Polido JC, Cermak SA. Physiological and behavioral stress and anxiety in children with autism spectrum disorders during routine oral care. *Biomed Res Int* 2014;2014:1–10. doi: [10.1155/2014/694876](https://doi.org/10.1155/2014/694876).
9. Marshall J, Sheller B, Williams BJ, Mancl L, Cowan C. Cooperation predictors for dental patients with autism. *Pediatr Dent* [Internet] 2007;29(5):369–76.
10. Dangulavanich W, Limsomwong P, Mittrakul K, Asvanund Y, Arunakul M. Factors associated with cooperative levels of Autism Spectrum Disorder children during dental treatments. *Eur J Paediatr Dent* [Internet] 2017;18(3):231–6. doi: [10.23804/ejpd.2017.18.03.11](https://doi.org/10.23804/ejpd.2017.18.03.11).
11. Nelson T, Chim A, Sheller BL, McKinney CM, Scott JM. Predicting successful dental examinations for children with autism spectrum disorder in the context of a dental desensitization program. *J Am Dent Assoc* [Internet] 2017;148(7):485–92. doi: [10.1016/j.adaj.2017.03.015](https://doi.org/10.1016/j.adaj.2017.03.015).
12. Stein LI, Polido JC, Cermak SA. Oral care and sensory over-responsivity in children with autism spectrum disorders. *Pediatr Dent* 2013;35(3):230–5. doi: [10.5014/ajot.2012.004085](https://doi.org/10.5014/ajot.2012.004085).
13. Lewis C, Vigo L, Novak L, Klein EJ. Listening to parents: a qualitative look at the dental and oral care experiences of children with autism spectrum disorder. *Pediatr Dent* [Internet] 2015;37(7):E98–104 PMID: 26883603.
14. da Silva SN, Gimenez T, Souza RC, et al. Oral health status of children and young adults with autism spectrum disorders: systematic review and meta-analysis. *Int J Paediatr Dent* 2017;27(5):388–98. doi: [10.1111/ipd.12274](https://doi.org/10.1111/ipd.12274).
15. Zhang Y, Lin L, Liu J, Shi L, Lu J. Dental caries status in autistic children: a meta-analysis. *J Autism Dev Disord* [Internet] 2020;50(4):1249–57. doi: [10.1007/s10803-019-04256-x](https://doi.org/10.1007/s10803-019-04256-x).
16. Pi X, Liu C, Li Z, Guo H, Jiang H, Du M. A meta-analysis of oral health status of children with autism. *J Clin Pediatr Dent* 2020;44(1):1–7. doi: [10.17796/1053-4625-44.1.1](https://doi.org/10.17796/1053-4625-44.1.1).
17. Hidayatullah T, Agustiani H, Setiawan AS. Behavior management-based applied behaviour analysis within dental examination of children with autism spectrum disorder. *Dental Journal (Majalah Kedokteran Gigi)* 2018;51(2):71. doi: [10.20473/j.djmk.v51.i2.p71-75](https://doi.org/10.20473/j.djmk.v51.i2.p71-75).
18. Orellana LM, Martínez-Sanchis S, Silvestre FJ. Training adults and children with an autism spectrum disorder to be compliant with a clinical dental assessment using a TEACCH-based approach. *J Autism Dev Disord* 2014;44(4):776–85. doi: [10.1007/s10803-013-1930-8](https://doi.org/10.1007/s10803-013-1930-8).

19. Zink AG, Diniz MB, Rodrigues dos Santos MTB, Guaré RO. Use of a Picture Exchange Communication System for preventive procedures in individuals with autism spectrum disorder: pilot study. *Special Care in Dentistry* [Internet] 2016;36(5):254–9. doi: [10.1111/scd.12183](https://doi.org/10.1111/scd.12183).
20. Cermak SA, Stein Duker LI, Williams ME, et al. Feasibility of a sensory-adapted dental environment for children with autism. *Am J Occup Ther* [Internet] 2015;69(3):6903220020p110. doi: [10.5014/ajot.2015.013714](https://doi.org/10.5014/ajot.2015.013714).
21. Virues-Ortega J, Julio FM, Pastor-Barriuso R. The TEACCH program for children and adults with autism: A meta-analysis of intervention studies. *Clin Psychol Rev* [Internet] 2013;33(8):940–53. doi: [10.1016/j.cpr.2013.07.005](https://doi.org/10.1016/j.cpr.2013.07.005).
22. Orellana LM, Cantero-Fuentealba C, Schmidlin-Espinoza L, Luengo L. Psychoeducational intervention to improve oral assessment in people with autism spectrum disorder, BIO-BIO region, Chile. *Med Oral Patol Oral Cir Bucal* [Internet] 2019;24(1):e37–46. doi: [10.4317/medoral.22560](https://doi.org/10.4317/medoral.22560).
23. Dahlan S. *Pintu Gerbang Memahami Epidemiologi Biostatistik dan Metodologi Penelitian*. 2 Ed. Jakarta: Epidemiologi Indonesia; 2017. p. 151.
24. Mathur J, Diwanji A, Sarvaiya B, Sharma D. Identifying Dental Anxiety in Children's Drawings and correlating It with Frankl \s Behavior Rating Scale. *Int J Clin Pediatr Dent* 2017;10(1):24–8. doi: [10.5005/jp-journals-10005-1401](https://doi.org/10.5005/jp-journals-10005-1401).
25. Heima M, Stehli K. Reliability and validity of the dental anxiety question used with children. *Anesth Prog* 2022;69(3):13–9. doi: [10.2344/anpr-69-01-04](https://doi.org/10.2344/anpr-69-01-04).
26. Klein U. Characteristics of patients with autistic disorder (AD) presenting for dental treatment: a survey and chart review. *Special Care in Dentistry* 1999;19(5):200–7. doi: [10.1111/j.1754-4505.1999.tb01386.x](https://doi.org/10.1111/j.1754-4505.1999.tb01386.x).
27. Whitehead G, Durr D, Adair S, Proskin H. Monitoring of sedated pediatric dental patient: behavior. *Am Soc Dent Chil* 1988;329–38.
28. Stein LI, Polido JC, Mailloux Z, Coleman GG, Cermak SA. Oral care and sensory sensitivities in children with autism spectrum disorders. *Spec Care Dentist* [Internet] 2011;31(3):102–10. doi: [10.1111/j.1754-4505.2011.00187.x](https://doi.org/10.1111/j.1754-4505.2011.00187.x).
29. Carter L, Harper JM, Luiselli JK. Dental desensitization for students with autism spectrum disorder through graduated exposure, reinforcement, and reinforcement-fading. *J Dev Phys Disabil* 2019;31(2):161–70. doi: [10.1007/s10882-018-9635-8](https://doi.org/10.1007/s10882-018-9635-8).
30. Mah JW, Tsang P. Visual schedule system in dental care for patients with autism: a pilot study. *J Clin Pediatr Dent* [Internet] 2016;40(5):393–9. doi: [10.17796/1053-4628-40.5.393](https://doi.org/10.17796/1053-4628-40.5.393).
31. Nissan S, Yochman A, Blumer S, Kharouba J, Peretz B. Children's Responses to Sensory Stimuli and their Behavior in the Dental Office. *J Clin Pediatr Dent* [Internet] 2017;41(1):10–7 Jan [cited 2019 Feb 16]. doi: [10.17796/1053-4628-41.1.10](https://doi.org/10.17796/1053-4628-41.1.10).
32. Mesibov GB, Shea V. The TEACCH program in the era of evidence-based practice. *J Autism Dev Disord* 2010;40(5):570–9. doi: [10.1007/s10803-009-0901-6](https://doi.org/10.1007/s10803-009-0901-6).
33. Panerai S, Ferrante L, Zingale M. Benefits of the Treatment and Education of Autistic and Communication Handicapped Children (TEACCH) programme as compared with a non-specific approach. *J Intellect Disabil Res* [Internet] 2002;46(Pt 4):318–27. doi: [10.1046/j.1365-2788.2002.00388.x](https://doi.org/10.1046/j.1365-2788.2002.00388.x).
34. McCoy K, Hermansen E. Video Modelling for individuals with Autism: Review of Model Types and Effects. *Children* 2011;30(4):183213. doi: [10.1353/etc.2007.0029](https://doi.org/10.1353/etc.2007.0029).
35. Alzyoudi M, Sartawi AA, Almuhiiri O. The impact of video modeling on improving social skills in children. *J Am Acad Special Edu Prof* [Internet] 2014:219–30. doi: [10.1111/1467-8578.12057](https://doi.org/10.1111/1467-8578.12057).
36. Stein LI, Polido JC, Cermak SA. Oral care and sensory concerns in autism. *Am J Occup Ther* [Internet] 2012;66(5):e73–6. doi: [10.5014/ajot.2012.004085](https://doi.org/10.5014/ajot.2012.004085).
37. Laurent AC, Gorman K. Development of emotion self-regulation among young children with autism spectrum disorders: the role of parents. *J Autism Dev Disord* 2018;48(4):1249–60. doi: [10.1007/s10803-017-3430-8](https://doi.org/10.1007/s10803-017-3430-8).
38. Thomas N, Blake S, Morris C, Moles DR. Autism and primary care dentistry: parents' experiences of taking children with autism or working diagnosis of autism for dental examinations. *Int J Paediatr Dent* 2018;28(2):22638. doi: [10.1111/ipd.12345](https://doi.org/10.1111/ipd.12345).
39. Columb MO, Atkinson MS. Statistical analysis: sample size and power estimations. *BJA Educ* 2016;16(5):15961. doi: [10.1093/bjaed/mkv034](https://doi.org/10.1093/bjaed/mkv034).
40. Murti Bhisma. *Design dan Ukuran Sampel untuk Penelitian Kuantitatif dan Kualitatif di Bidang Kesehatan*. Yogyakarta: Gadjah Mada University Press; 2016. p. 112–28.
41. Deeks JJ, Dinnes J, D'amico R. Evaluating non-randomised intervention studies HTA Health Technology Assessment NHS R&D HTA Programme [Internet], 7. Health Technology Assessment; 2003. doi: [10.3310/hta7270](https://doi.org/10.3310/hta7270).